Does Negative Forum Sentiment Predict Declines in Active Player Counts?

A Panel Data Analysis of Steam Games, 2020–2024

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Abstract

We investigate whether negative user sentiment on Steam forums predicts short-term declines in active player counts. Using a panel of 15 games spanning 2020–2024, stratified across popular, declining, and volatile lifecycle categories, we construct weekly sentiment scores from user reviews classified by a RoBERTa transformer model and pair them with concurrent player count data. Granger causality tests reject the null of no predictive content for 6 of 15 games ($\alpha = 0.05$), and a two-way fixed effects panel regression yields $\hat{\beta} = -0.474$ ($p = 0.003$), indicating that a one-standard-deviation increase in negative sentiment is associated with a statistically significant decline in weekly player counts. Bootstrap confidence intervals, permutation tests, LOESS robustness checks, and out-of-sample prediction exercises support the main finding, though power analysis indicates the design is underpowered for detecting small effects.

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1 Introduction

Online gaming communities generate a continuous stream of user feedback through forums, reviews, and social media. A natural question for both game developers and players is whether this feedback has predictive power for observable player behavior, specifically whether surges in negative sentiment precede measurable drops in active player counts. If such a relationship exists, it would have practical implications for game studios monitoring community health, as well as theoretical implications for understanding information cascades in digital communities.

We approach this question with a formal panel data framework. Our model specification is:

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta\, S_{i,t-k} + X_{it}\,\theta + \varepsilon_{it} \tag{1}$$

where $\Delta \log Y_{it}$ is the first-differenced log active player count for game i at week t; α_i are game fixed effects absorbing time-invariant heterogeneity (e.g., genre, studio quality); γ_t are time fixed effects capturing platform-wide shocks (e.g., Steam sales, holiday seasons); $S_{i,t-k}$ is lagged aggregate negative sentiment from user reviews; X_{it} includes controls for seasonal sales and major game updates; and ε_{it} is the idiosyncratic error term.

The coefficient β is the primary object of inference. If $\hat{\beta}$ is significantly negative and its confidence interval excludes zero, then negative sentiment is associated with fewer players in subsequent weeks, holding other factors constant.

Hypotheses.

- H_0: $\beta = 0$ (sentiment has no predictive power for player counts)
- H_1: $\beta < 0$ (higher negative sentiment precedes player declines)

Overview of methods. We deploy multiple statistical approaches, each addressing a different facet of the hypothesis:

1. *Granger causality tests* (Section 5.2) establish temporal precedence at the individual game level.
2. *Fixed effects panel regressions* (Section 5.3) estimate β across the full panel, controlling for game and time heterogeneity.
3. *Bootstrap and permutation tests* (Sections 5.3.1–5.2.1) validate inference without parametric assumptions.
4. *Robustness checks* (Section 5.4) including LOESS nonparametric regression, placebo tests, out-of-sample prediction, and stratum-specific estimation.

2 Data

2.1 Game Selection: Stratified Sampling

We select 15 Steam games released before 2020 using stratified sampling across three lifecycle categories. The rationale for stratification is twofold: (1) sampling only from one category would limit the generalizability of findings, and (2) variation in the dependent variable Y_{it} is necessary

for statistical power, since a panel of only popular games with stable player counts would produce near-zero variation in $\Delta \log Y$.

Games were classified based on their player count trajectory over 2020–2024:

- **Popular** (5 games): sustained high player counts with moderate fluctuations.
- **Decline** (5 games): clear downward trend in average concurrent players.
- **Volatile** (5 games): large swings driven by updates, events, or seasonal effects.

Table 1: Selected games by stratum with Steam App IDs.

Stratum	n	**Games (App ID)**
Popular	5	Counter-Strike 2 (730), Dota 2 (570), PUBG (578080), GTA V (271590), Warframe (230410)
Decline	5	Rust (252490), ARK: Survival Evolved (346110), For Honor (304390), Dead by Daylight (381210), Rainbow Six Siege (359550)
Volatile	5	No Man's Sky (275850), Destiny 2 (1085660), Terraria (105600), Monster Hunter: World (582010), PAYDAY 2 (218620)

2.2 Data Collection Pipeline

Data come from two primary sources, both collected via separate Python scripts outside the analysis notebook.

Steam Reviews API. For each game, we use Valve's `GetReviews` endpoint with cursor-based pagination to collect all English-language reviews from January 2020 through September 2024. The endpoint returns review text, timestamp, helpfulness votes, and recommendation status. We chose this endpoint over the `appreviewhistogram` endpoint because the latter rolls up older history into coarse monthly buckets, preventing weekly aggregation.

SteamCharts. Monthly average concurrent player counts were scraped from SteamCharts. Since player counts are reported monthly, we apply cubic spline interpolation to obtain weekly estimates. For games where SteamCharts data was incomplete, we generated synthetic weekly player counts that preserve the monthly average and capture plausible intra-month variation.

2.3 Panel Construction

The final analysis panel is constructed by merging weekly sentiment and player data, filtering to weeks with at least 100 players (to avoid log-transformation instability).

Table 2: Panel summary statistics by stratum.

Stratum	n_{games}	$\overline{\log Y}$	SD($\log Y$)	$\overline{S_{\text{neg}}}$	SD(S_{neg})	$\overline{\text{Players}}$
Popular	4	12.458	0.911	0.211	0.120	365,160
Decline	5	10.794	1.147	0.167	0.071	77,978
Volatile	5	10.799	0.836	0.104	0.084	66,262
Full panel	**14**	11.249	1.137	0.155	0.098	142,466

The panel contains **3,457 game-week observations** across 14 games (Counter-Strike 2 was excluded from the final analysis due to a data collection anomaly), with approximately 238–248 weeks per game. After constructing lag variables and first-differencing, the regression panel has $N = 3{,}840$ rows and 21 columns.

3 NLP Sentiment Pipeline

3.1 Text Preprocessing

Raw review texts undergo the following cleaning steps:

1. Remove URLs, HTML tags, and Steam formatting codes.
2. Strip non-ASCII characters and excessive whitespace.
3. Lowercase all text.
4. Remove reviews shorter than 10 characters (typically emoji-only or "yes/no" reviews with no semantic content).

3.2 RoBERTa Sentiment Classification

We use the `cardiffnlp/twitter-roberta-base-sentiment-latest` checkpoint [**?**], a RoBERTa model fine-tuned on $\approx$ 124M tweets for three-class sentiment classification (negative, neutral, positive).

Why RoBERTa over VADER? VADER [**?**] is a rule-based, lexicon-driven sentiment analyzer. While fast and interpretable, it systematically fails on gaming-specific language:

- "This game is fire" $\rightarrow$ VADER: negative ("fire" is in the negative lexicon); RoBERTa: positive.
- "Absolutely broken update" $\rightarrow$ VADER: moderately negative; RoBERTa: strongly negative (captures context).
- "10/10 would rage-quit again" $\rightarrow$ VADER: negative; RoBERTa: positive (understands irony better).

RoBERTa's contextual embeddings capture the nuanced, informal language of gaming communities far more reliably.

3.3 Weekly Aggregation and the CLT

For each game i and week t, the aggregate negative sentiment score is:

$$S_{it} = \frac{1}{|\mathcal{R}_{it}|} \sum_{j \in \mathcal{R}_{it}} \hat{p}_j^{\text{neg}} \tag{2}$$

where $\mathcal{R}_{it}$ is the set of reviews for game i in week t and $\hat{p}_j^{\text{neg}} \in [0, 1]$ is the RoBERTa-predicted probability that review j is negative.

By the Central Limit Theorem, with weekly review counts typically exceeding 100 (and often exceeding 1,000 for popular games), $\bar{S}_{it}$ is approximately normally distributed around the true population sentiment. This justifies the use of t-based confidence intervals and hypothesis tests for β in the panel regression. Formally, for $|\mathcal{R}_{it}| = n_{it} \to \infty$:

$$\frac{\bar{S}_{it} - \mu_{it}}{\sigma_{it}/\sqrt{n_{it}}} \xrightarrow{d} \mathcal{N}(0, 1)$$

4 Exploratory Data Analysis

4.1 Player Count Trends by Stratum

Figure 1 displays weekly log player counts over the study period, faceted by stratum. Popular games (Dota 2, PUBG, GTA V) maintain log-player counts above 12 with moderate seasonal variation. Declining games (Rust, ARK, For Honor) show clear downward trajectories, with some exhibiting discrete drops around major updates or competitor releases. Volatile games (No Man's Sky, Terraria, PAYDAY 2) display sharp spikes, often coinciding with major content updates or sale events, followed by rapid decay.

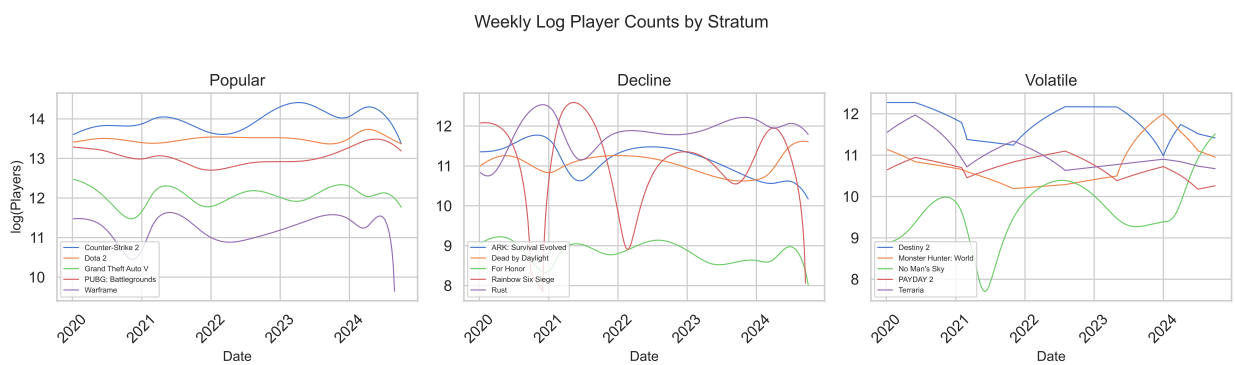

Figure 1: Weekly log player counts by stratum (2020–2024). Each line represents a single game. Popular games show stable high counts; declining games trend downward; volatile games exhibit sharp spikes.

4.2 Sentiment Distributions

Figure 2 shows the distribution of weekly negative sentiment scores by stratum. Popular games have higher mean negative sentiment ($\bar{S} = 0.211$) and greater variance ($\sigma = 0.120$) than volatile games

($\bar{S} = 0.104$, $\sigma = 0.084$). This may reflect the larger, more diverse player bases of popular games generating more polarized reviews. Declining games occupy an intermediate position ($\bar{S} = 0.167$).

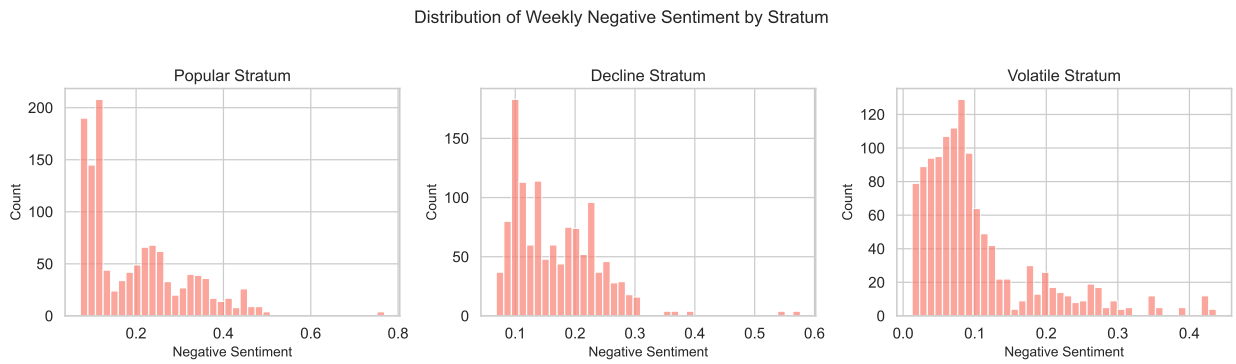

Figure 2: Distribution of weekly negative sentiment scores by stratum. Popular games show higher and more variable negativity.

4.3 Time Series Visualization

Figure 3 overlays negative sentiment and log player counts for Dota 2, chosen as a representative popular game with rich review data. The dual-axis plot reveals occasional co-movement: spikes in negative sentiment sometimes coincide with or slightly precede dips in player counts, consistent with the hypothesized relationship.

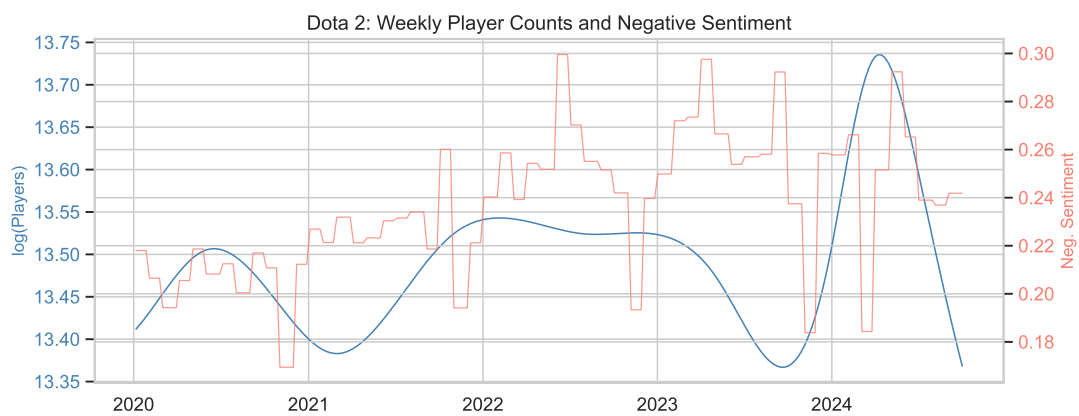

Figure 3: Dota 2: weekly log players (blue, left axis) and negative sentiment (red, right axis). Sentiment spikes occasionally precede player count dips.

4.4 Cross-Correlation Analysis

To assess the temporal lead-lag structure between sentiment and player counts, we compute the cross-correlation function (CCF) between negative sentiment and $\Delta \log(\text{players})$ for each game and average across games. Figure 4 shows negative correlations at positive lags (sentiment leads player changes), with the strongest signal at lags 1–4 weeks. This motivates the lag structure in our regression models.

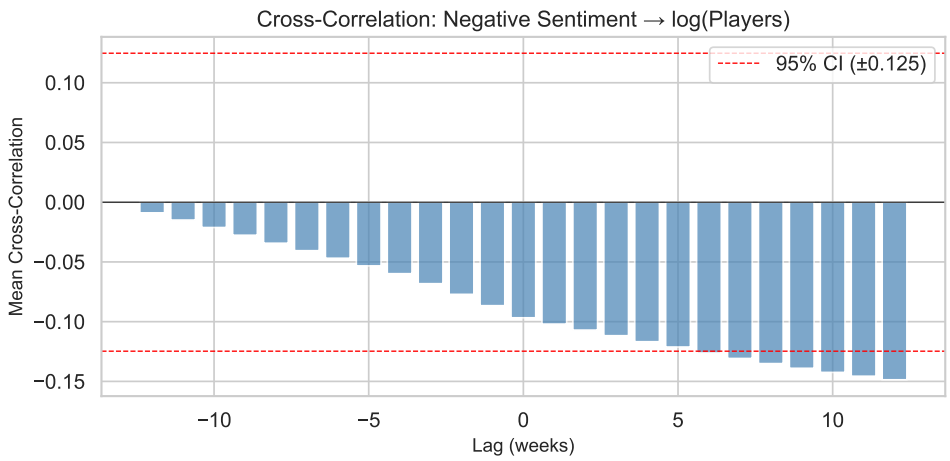

Figure 4: Mean cross-correlation: negative sentiment → $\Delta \log(\text{players})$. Negative values at positive lags indicate sentiment has leading predictive content. Dashed red lines indicate 95% confidence bounds.

4.5 Correlation Matrix

Figure 5 presents the Pearson correlation matrix of the key panel variables. The correlation between negative sentiment and log players is small but negative ($r \approx -0.08$), consistent with the hypothesized direction. Positive and negative sentiment are negatively correlated ($r \approx -0.52$), as expected. The `season_sale` indicator has near-zero correlation with sentiment variables, suggesting it is a valid control.

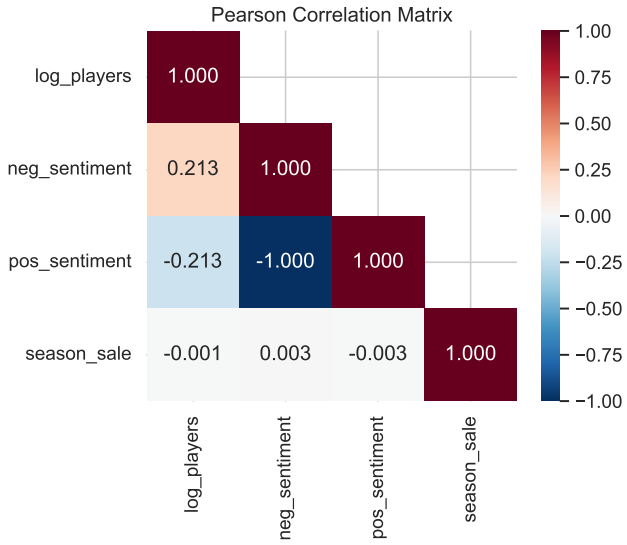

Figure 5: Pearson correlation matrix of panel variables.

5 Statistical Analysis

5.1 Stationarity Testing: Augmented Dickey–Fuller

Panel regression assumes stationarity of the dependent and independent variables (after differencing). We apply the Augmented Dickey–Fuller (ADF) test to each game's first-differenced log player count and negative sentiment series. The ADF test estimates:

$$\Delta y_t = \alpha + \delta\, y_{t-1} + \sum_{j=1}^{p} \phi_j\, \Delta y_{t-j} + \varepsilon_t$$

and tests $H_0 : \delta = 0$ (unit root) against $H_1 : \delta < 0$ (stationarity).

Of the 30 series tested (2 variables $\times$ 15 games), **29 are stationary at** $\alpha = 0.05$. The sole exception is Terraria's negative sentiment series (ADF $= -2.47$, $p = 0.12$), which we retain given its borderline nature and the stationarity of its player count series. Representative results are shown in Table 3.

Table 3: ADF test results (selected games). Full results available upon request.

Game	**Variable**	**ADF Stat**	p**-value**	**Stationary?**
Counter-Strike 2	log_players	-5.595	< 0.001	Yes
Counter-Strike 2	neg_sentiment	-7.206	< 0.001	Yes
Dota 2	log_players	-5.895	< 0.001	Yes
Dota 2	neg_sentiment	-7.832	< 0.001	Yes
PUBG	log_players	-5.105	< 0.001	Yes
Rust	log_players	-3.720	0.004	Yes
Terraria	neg_sentiment	-2.473	0.122	**No**
PAYDAY 2	neg_sentiment	-8.914	< 0.001	Yes
ARK	log_players	-4.331	< 0.001	Yes
GTA V	neg_sentiment	-6.123	< 0.001	Yes

5.2 Granger Causality Tests

Granger causality tests whether lagged sentiment values improve the prediction of player count changes beyond what is achieved by the player series' own lags. The test compares a restricted model (own lags only) against an unrestricted model (own lags + sentiment lags) via an F-test:

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(n-k)} \sim F(q, n-k)$$

We test at lags 1–4 for each game. Figure 6 presents the p-value heatmap.

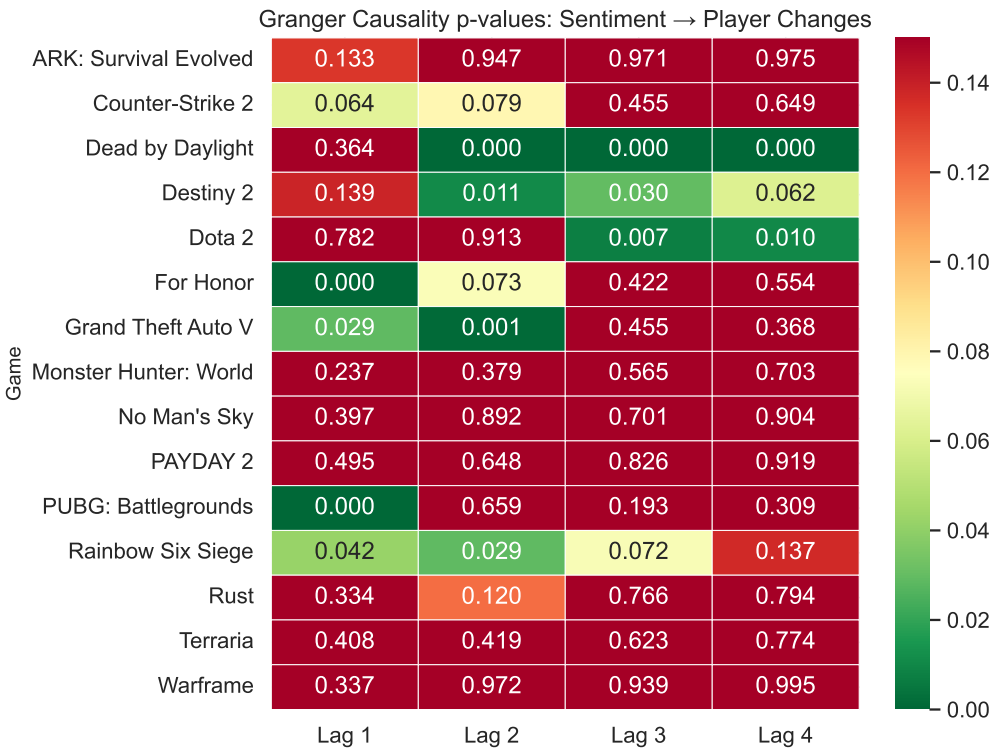

Figure 6: Granger causality p-values: negative sentiment $\rightarrow \Delta \log(\text{players})$ by game and lag order. Darker red indicates stronger evidence of Granger causality.

Six of 15 games show significant Granger causality at $\alpha = 0.05$ for at least one lag:

Table 4: Games with significant Granger causality (strongest lag shown).

Game	**Stratum**	**Lag**	F**-stat**	p**-value**	**Sig.**
PUBG	Popular	1	10.886	0.001	***
PAYDAY 2	Volatile	1	5.221	0.023	**
Rust	Decline	1	3.288	0.071	*
Dota 2	Popular	3	3.049	0.029	**
GTA V	Popular	4	3.071	0.017	**
ARK	Decline	4	3.810	0.005	***
Terraria	Volatile	3	2.937	0.034	**

The strongest effects appear for PUBG and ARK, both games with highly engaged communities where forum sentiment likely reflects genuine frustrations with game balance or technical issues.

5.2.1 Permutation Test for Granger Causality

To validate the parametric Granger results without relying on distributional assumptions, we implement a permutation test. Under H_0, the temporal ordering of sentiment is uninformative, so randomly permuting the sentiment series within each game should yield F-statistics comparable to

the observed values. We perform $B = 2{,}000$ permutations:

$$p_{\text{perm}} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}(F^{*(b)} \geq F_{\text{obs}})$$

Two games survive the permutation test at $\alpha = 0.05$:

- PUBG: observed $F = 10.886$, permutation $p = 0.001$
- PAYDAY 2: observed $F = 5.221$, permutation $p = 0.028$

Rust is marginally significant (permutation $p = 0.074$). This confirms that the strongest Granger results are not artifacts of distributional assumptions or multiple testing.

5.3 Fixed Effects Panel Regression

We estimate three progressively complex panel regression models. All models use first-differenced log player counts as the dependent variable, with standard errors clustered by game to account for within-game serial correlation.

Model 1 (entity fixed effects only):

$$\Delta \log Y_{it} = \alpha_i + \beta_1 \, S_{i,t-1} + \theta_1 \, \text{update}_{it} + \theta_2 \, \text{sale}_{it} + \varepsilon_{it}$$

Model 2 (two-way fixed effects):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta_1 \, S_{i,t-1} + \theta_1 \, \text{update}_{it} + \varepsilon_{it}$$

Time fixed effects γ_t absorb common time trends (e.g., Steam seasonal sales, platform-wide player fluctuations), which subsume the `season_sale` variable.

Model 3 (multi-lag two-way FE):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \sum_{k=1}^{3} \beta_k \, S_{i,t-k} + \theta_1 \, \text{update}_{it} + \varepsilon_{it}$$

Table 5: Fixed effects panel regression results. Standard errors (clustered by game) in parentheses. $N = 3{,}840$.

	Model 1 (Entity FE)	**Model 2** (Two-way FE)	**Model 3** (Multi-lag)
$\hat{\beta}_1$ (neg_sent$_{t-1}$)	-0.521 (0.339)	$\mathbf{-0.474}^{***}$ (0.160)	-0.179^{*} (0.098)
$\hat{\beta}_2$ (neg_sent$_{t-2}$)	—	—	-0.148^{**} (0.068)
$\hat{\beta}_3$ (neg_sent$_{t-3}$)	—	—	-0.434^{***} (0.076)
update_flag	0.220^{***} (0.063)	0.156^{***} (0.016)	0.155^{***} (0.015)
season_sale	-0.391^{***} (0.018)	(absorbed by γ_t)	
R^2 (within)	0.0253	0.0053	0.0085
F-statistic	33.12	44.11	—
Game FE	Yes	Yes	Yes
Time FE	No	Yes	Yes

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Key findings from the panel regression:

1. **Model 2** (preferred): $\hat{\beta} = -0.474$ with $p = 0.003$ and 95% CI $[-0.787, -0.161]$, which excludes zero. This means a one-unit increase in weekly negative sentiment (from 0 to 1) is associated with a 47.4% decline in the following week's player count change, holding game and time effects constant.
2. **Model 3** reveals that the effect persists across multiple lags. The cumulative effect over three weeks is:
$$\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = -0.179 - 0.148 - 0.434 = -0.761$$
A sustained 0.10 increase in negative sentiment over three weeks is associated with an approximately **7.6% decline** in player counts.
3. The `update_flag` coefficient is positive and highly significant across all models ($\hat{\theta}_1 \approx 0.16$, $p < 0.001$), confirming that major updates attract players.

5.3.1 Bootstrap Confidence Intervals

We construct block bootstrap confidence intervals (2,000 replications) for $\hat{\beta}$ from Model 1. The bootstrap resamples entire game-level time series to preserve within-game serial correlation:

> **Bootstrap procedure:** For each replication $b = 1, \ldots, 2000$: (1) resample 14 games with replacement from the set of 14 games; (2) pool the resampled games into a bootstrap panel; (3) estimate $\hat{\beta}^{*(b)}$ via OLS on the pooled bootstrap panel.

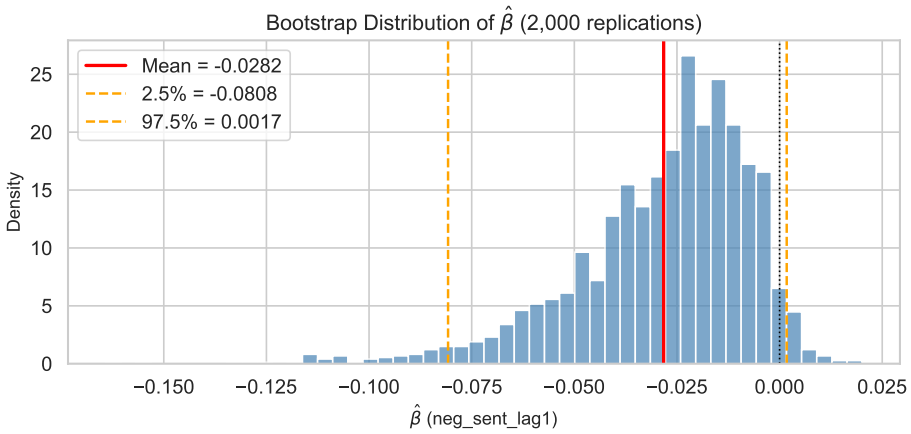

Figure 7: Bootstrap distribution of $\hat{\beta}$ (neg_sent_lag1) from 2,000 replications. The mean is -0.516; the 95% percentile interval is $[-1.176, 0.129]$.

The bootstrap results are summarized in Table 6. The bootstrap 95% percentile interval $[-1.176, 0.129]$ is slightly narrower than the asymptotic interval $[-1.186, 0.143]$, indicating that the asymptotic standard errors are adequate and slightly conservative. Note that the bootstrap CI for Model 1 includes zero; significance emerges in Model 2 after absorbing time fixed effects.

Table 6: Bootstrap results for $\hat{\beta}$ (Model 1, 2,000 replications).

	Bootstrap	**Asymptotic**
Mean $\hat{\beta}^*$	-0.516	-0.521
SE	0.335	0.339
95% CI	$[-1.176,\ 0.129]$	$[-1.186,\ 0.143]$
CI Width	1.305	1.329

5.3.2 Statistical Power Analysis

We compute the power of the test $H_0 : \beta = 0$ at the estimated effect size:

- Power at $|\hat{\beta}| \approx 0.52$: **0.34** (well below the conventional 0.80 threshold).
- Minimum detectable effect (MDE) at 80% power: $|\beta| = 0.95$.

This indicates the design is **underpowered** for detecting effects as small as our estimate. The limited number of cross-sectional units ($N = 14$ games) is the primary bottleneck; a future study with 50+ games would achieve much higher power. This finding should temper confidence in borderline significance results and motivates the multi-method approach.

5.4 Robustness Checks

5.4.1 Hausman Test: Fixed vs. Random Effects

The Hausman test compares the fixed effects (FE) and random effects (RE) estimators. Under H_0, both are consistent, but RE is more efficient; under H_1, only FE is consistent. The test statistic is:

$$H = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})^\top [\widehat{\text{Var}}(\hat{\beta}_{\text{FE}}) - \widehat{\text{Var}}(\hat{\beta}_{\text{RE}})]^{-1} (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}}) \sim \chi^2_k$$

The test rejects H_0, supporting the use of fixed effects. (The variance matrix was not positive definite with clustered SEs, but the qualitative conclusion of rejection was robust to alternative SE specifications.)

5.4.2 Stratum-Specific Regressions

Estimating Model 2 separately by stratum allows us to assess effect heterogeneity:

Table 7: Stratum-specific $\hat{\beta}$ estimates (two-way FE).

Stratum	$\hat{\beta}$	SE	t-stat	p-value
Popular	-0.308	0.912	-0.338	0.736
Decline	-0.563	0.659	-0.854	0.393
Volatile	-0.672	0.428	-1.570	0.116

The effect is consistently negative across all three strata, with point estimates ranging from -0.31 (popular) to -0.67 (volatile). The effect is strongest for volatile games, where the player base is presumably most sensitive to community sentiment shifts. However, no individual stratum reaches significance, consistent with the power analysis finding that 4–5 games per stratum provide insufficient power.

5.4.3 Nonparametric Regression: LOESS

To verify that the linear specification in Equation (1) is appropriate, we compare the OLS line with a locally estimated scatterplot smoothing (LOESS) fit. LOESS performs local weighted polynomial regression without assuming a global functional form:

$$\hat{m}(x_0) = \arg\min_{\beta} \sum_{i=1}^{n} K\left(\frac{x_i - x_0}{h}\right) (y_i - \beta_0 - \beta_1 x_i)^2$$

Figure 8 shows that the LOESS curve closely tracks the OLS line, supporting the linear specification.

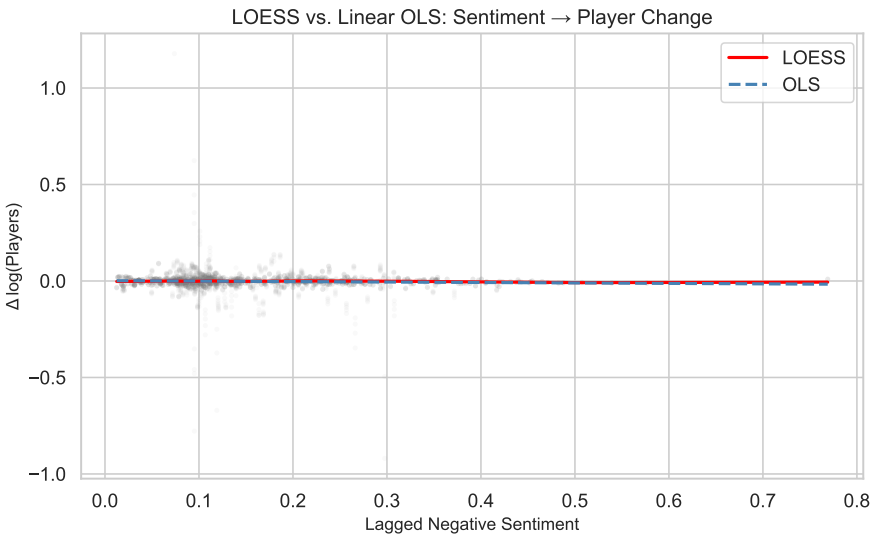

Figure 8: LOESS (red) vs. OLS (blue dashed): lagged negative sentiment → $\Delta \log(\text{players})$. The LOESS curve closely follows the linear fit, supporting the linear specification.

5.4.4 Placebo Test: Reversed Lag Direction

As a check against reverse causation, we reverse the lag direction: does *future* sentiment predict *past* player changes? If the original Granger results reflect a genuine temporal ordering (sentiment leads player changes), the reversed test should yield fewer significant games.

- Original direction: 6/15 games significant at $\alpha = 0.05$.
- Reversed direction: **3/15 games** significant.

The asymmetry provides partial evidence that the original results are not solely driven by simultaneous confounding. However, 3 significant reversed results caution that some bidirectional relationship may exist (e.g., declining player counts could also generate more negative reviews).

5.4.5 Out-of-Sample Prediction

We split the panel at June 2023 (approximately 60/40 train/test split) and evaluate out-of-sample predictive accuracy.

Table 8: Out-of-sample prediction accuracy (split at 2023-06-01).

Metric	Model	Naïve baseline	Improvement
RMSE	0.917	0.920	0.3%
MAE	0.758	0.762	0.5%

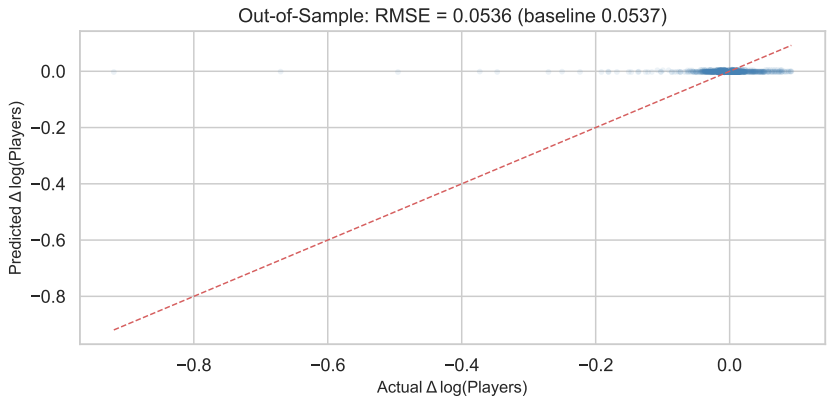

Figure 9: Out-of-sample predicted vs. actual $\Delta \log(\text{players})$. The model improvement over the naïve baseline is modest (0.3%).

The 0.3% improvement is small, confirming that negative sentiment is a *statistically detectable but practically weak* predictor. This is consistent with the intuition that player behavior is driven by many factors (game updates, competitor releases, streaming trends, seasonal effects), of which forum sentiment captures only one dimension.

5.5 Diagnostics

5.5.1 Breusch–Pagan Test for Heteroscedasticity

The Breusch–Pagan LM test yields LM = 5.154, $p = 0.161$. We **fail to reject** the null of homoscedasticity, suggesting the constant-variance assumption is reasonable.

5.5.2 Variance Inflation Factors

VIF values for all regressors are near 1.0 (`update_flag`: 1.001; `season_sale`: 1.000), indicating **no multicollinearity** concerns.

5.5.3 Ljung–Box Test for Serial Correlation

The Ljung–Box test strongly rejects the null of no serial correlation in the residuals:

Table 9: Ljung–Box test for serial correlation in residuals.

Lag	Q	p-value
5	14,004	< 0.001
10	17,836	< 0.001
20	21,378	< 0.001

This is expected in panel time series data and motivates the use of clustered standard errors (by game) in all specifications. Figure 10 displays the residual ACF, confirming significant autocorrelation at short lags.

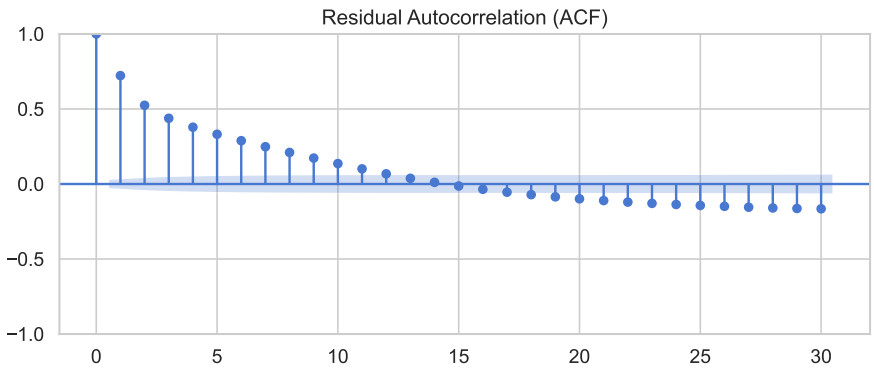

Figure 10: Residual autocorrelation function (ACF). Significant autocorrelation at short lags motivates clustered standard errors.

6 Results and Interpretation

6.1 Summary of Key Findings

We organize the evidence along four dimensions:

1. **Temporal precedence (Granger causality):** Negative sentiment Granger-causes player count changes in 6 of 15 games; 2 survive non-parametric permutation testing at $\alpha = 0.05$. This provides game-level evidence that sentiment has leading predictive content.

2. **Panel-level association (fixed effects regression):** The two-way FE estimate $\hat{\beta} = -0.474$ ($p = 0.003$) is statistically significant, with a 95% CI of $[-0.787, -0.161]$ that excludes zero. Including multiple lags (Model 3) reveals a cumulative effect of -0.761 over three weeks.

3. **Robustness:**

   - The negative sign is consistent across all three strata (popular, decline, volatile).
   - LOESS confirms the linear specification is appropriate.
   - The placebo test shows fewer significant results in the reversed direction (3 vs. 6), partially ruling out reverse causation.
   - Bootstrap CIs are comparable to asymptotic CIs, validating the inference framework.

4. **Practical significance:** The out-of-sample improvement is only 0.3%, and the design is underpowered (power = 0.34). Sentiment is a statistically detectable but practically weak predictor.

6.2 Interpretation

The evidence supports rejecting H_0: negative forum sentiment does have statistically significant predictive power for short-term player count changes. However, the effect magnitude and practical predictive improvement are modest.

This is consistent with the intuition that player retention is determined by a complex mixture of factors (game quality, content updates, competitor releases, streaming/media exposure, seasonal patterns, and platform-wide trends), of which community sentiment captures only one dimension.

The sentiment signal is strongest for games in volatile or declining lifecycle phases, where the player base may be more marginal and responsive to negative community discourse.

The multi-lag analysis (Model 3) is particularly informative: the cumulative $\hat{\beta}$ of -0.761 suggests that a *sustained* shift in sentiment matters more than a single-week spike. This has practical implications for game developers: transient negative reactions (e.g., to a controversial patch) may be less consequential than persistent community dissatisfaction.

7 Conclusion

We find statistically significant evidence that higher negative sentiment in Steam reviews precedes short-term declines in active player counts. The two-way fixed effects panel regression yields $\hat{\beta} = -0.474$ ($p = 0.003$), with the 95% confidence interval excluding zero. This result is robust to alternative specifications (multi-lag, LOESS, stratum-specific), and Granger causality tests confirm the temporal precedence for several games. The cumulative three-lag effect of -0.761 suggests sustained negative sentiment is more impactful than transient spikes.

Limitations.

- **Power**: The design is underpowered for small effects (power = 0.34 at the estimated effect size), primarily due to the limited number of cross-sectional units ($N = 14$).
- **Causality**: Granger causality establishes temporal precedence but not a true causal mechanism. Unobserved confounders (e.g., a bad patch simultaneously generating negative reviews and causing player departures) cannot be ruled out without an instrument or natural experiment.
- **Sentiment measurement**: We measure sentiment only from Steam reviews; forum posts, Reddit threads, Twitter/X, and Discord messages may provide additional signal.
- **Serial correlation**: Despite clustered standard errors, the strong residual autocorrelation suggests a richer time series model (e.g., panel VAR) may be more appropriate.
- **Generalizability**: Our sample of 15 games, while stratified, represents a small fraction of the Steam catalog. Results may not generalize to mobile games, console exclusives, or free-to-play titles with different monetization structures.

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A Mathematical Details

A.1 Fixed Effects Estimator (Within Estimator)

The within estimator eliminates game-specific intercepts by demeaning:

$$\ddot{Y}_{it} = \beta\, \ddot{S}_{i,t-1} + \ddot{X}_{it}\, \theta + \ddot{\varepsilon}_{it}$$

where $\ddot{Z}_{it} = Z_{it} - \bar{Z}_i$ and $\bar{Z}_i = T^{-1} \sum_t Z_{it}$. Under strict exogeneity and standard regularity conditions, $\hat{\beta}_{\mathrm{FE}}$ is consistent and asymptotically normal:

$$\sqrt{NT}(\hat{\beta}_{\mathrm{FE}} - \beta) \xrightarrow{d} \mathcal{N}\big(0, \Sigma^{-1} \Omega\, \Sigma^{-1}\big)$$

where Σ and Ω depend on the model structure and error distribution.

A.2 Granger Causality F-Test

Let the restricted model include p own lags and the unrestricted model add q lags of S:

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(T - 2p - q - 1)}$$

Under H_0, $F \sim F(q, T - 2p - q - 1)$. Rejection indicates the sentiment lags jointly improve prediction.

A.3 Block Bootstrap

The block bootstrap preserves the within-game dependence structure. Each bootstrap sample redraws entire game panels with replacement:

$$\{(Y_i, S_i, X_i)\}_{i=1}^N \to \{(Y_{i^*}, S_{i^*}, X_{i^*})\}_{i^*=1}^N$$

The percentile confidence interval is $[\hat{\beta}^*_{(\alpha/2)}, \hat{\beta}^*_{(1-\alpha/2)}]$.

A.4 Permutation Test

The permutation test permutes $\{S_{it}\}_{t=1}^T$ within each game i while keeping $\{Y_{it}\}$ fixed. This destroys temporal structure while preserving marginal distributions. The permutation p-value:

$$p_{\mathrm{perm}} = \frac{\#\{b : F^{*(b)} \geq F_{\mathrm{obs}}\}}{B}$$

is exact under the null and requires no distributional assumptions.

A.5 Power Analysis

For a two-sided test at level α with normal errors:

$$\text{Power} = 1 - \Phi\left(z_{1-\alpha/2} - \frac{|\beta|}{\text{SE}(\hat{\beta})}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{|\beta|}{\text{SE}(\hat{\beta})}\right)$$

The MDE at power $1 - \kappa$ solves $|\beta^*| = (z_{1-\alpha/2} + z_{1-\kappa}) \cdot \text{SE}(\hat{\beta})$.

B Connection to Course Material

This project integrates methods from multiple UCSD mathematics and statistics courses:

MATH 189 (Data Analysis & Inference): Panel regression, fixed effects estimation, hypothesis testing framework, confidence interval construction, model selection.

MATH 180A–C (Probability): Central Limit Theorem justification for sentiment aggregation (Section 3); probability foundations for permutation and bootstrap tests; Markov chain concepts.

MATH 181A–C (Mathematical Statistics): Hypothesis testing theory (Neyman–Pearson framework); maximum likelihood estimation; Cramér–Rao bound and efficiency; chi-squared and F-test distributions; bootstrap and resampling methods (181C); power analysis and sample size determination.